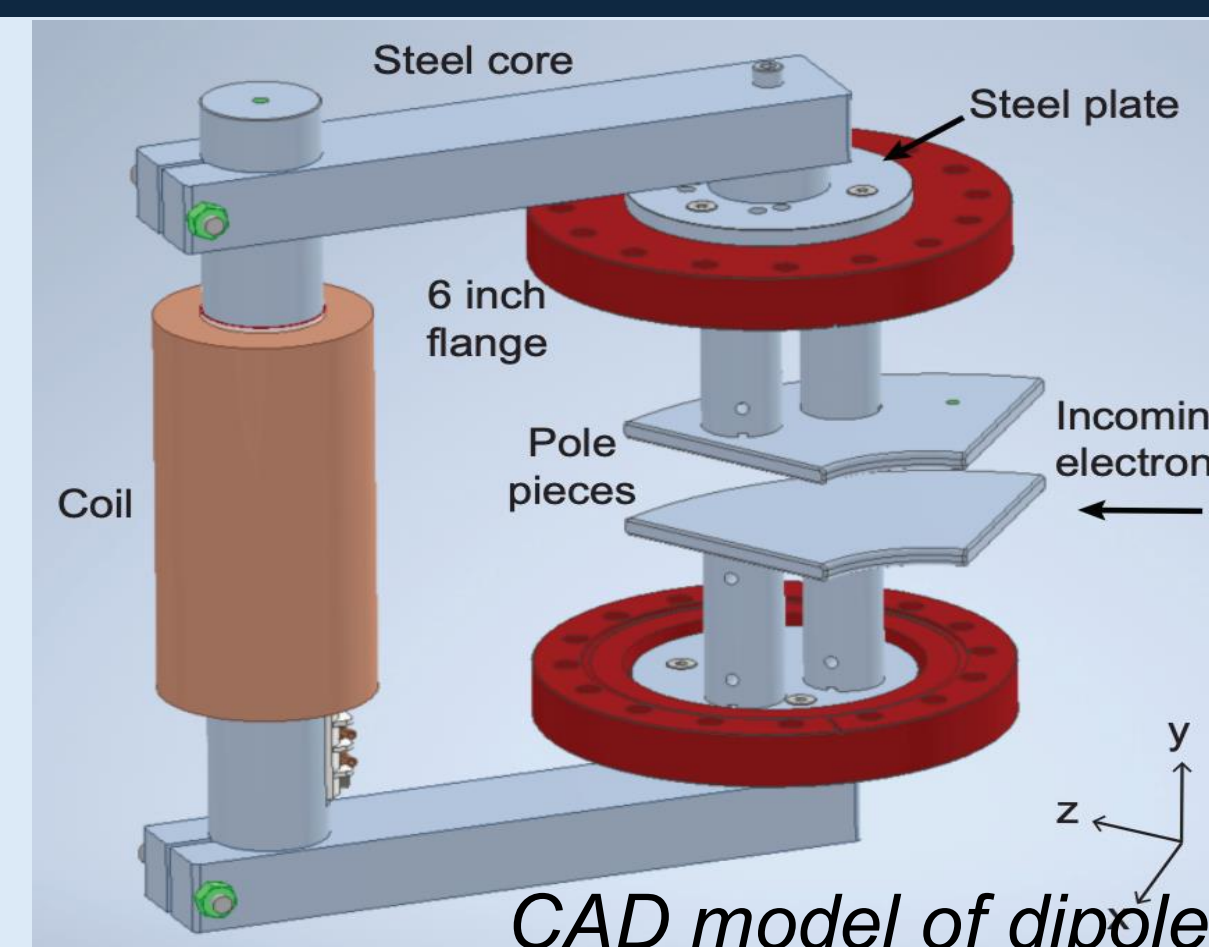


Background

The detection chamber is a crucial part of accelerator systems to characterise the beam result. Our linear accelerator uses a dipole magnet energy spectrometer to create an energy spread of the electron beam and bend it towards detectors.



The dipole is a bending magnet with two parallel pole pieces, driven by a coil around a steel core. An ideal dipole creates a uniform B field between the pole pieces, pointing perpendicular to the face of the pole piece.

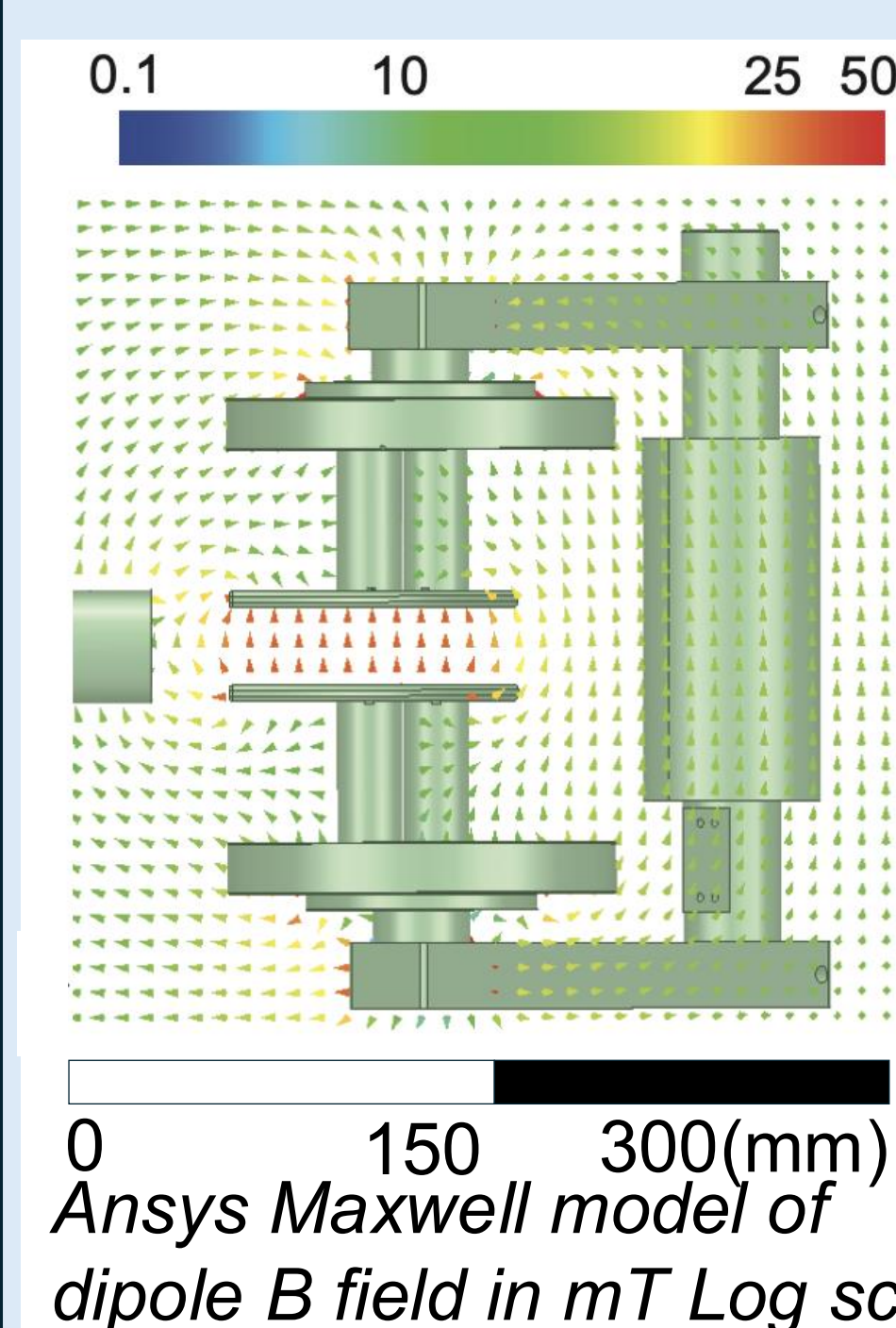
$$F = q(E + v \times B) = q(v \times B)$$

A charged particle entering the gap between pole pieces experiences a Lorentz force perpendicular to the magnetic field and its direction of travel.

The magnetic field (B) of the dipole bends the trajectory of the beam. The bending radius is based on electron energy, and is defined by the following equation:

$$\rho = \frac{mc\gamma\beta}{qB}$$

where γ is the Lorentz factor, β is the ratio of v to c , m is the mass, c is the speed of light, q is the charge, and B is the magnetic field. Particles with higher energy have a larger bending radius, so they follow a shallower trajectory.



The dipole uses a small B field to steer the sub-relativistic beam. While the B field is within the detection range of commercial high accuracy Hall effect probes, this projects aim to build a low-cost setup with hobbyist sensor chips that can provide a full scan of the B field between the two dipole pole pieces.

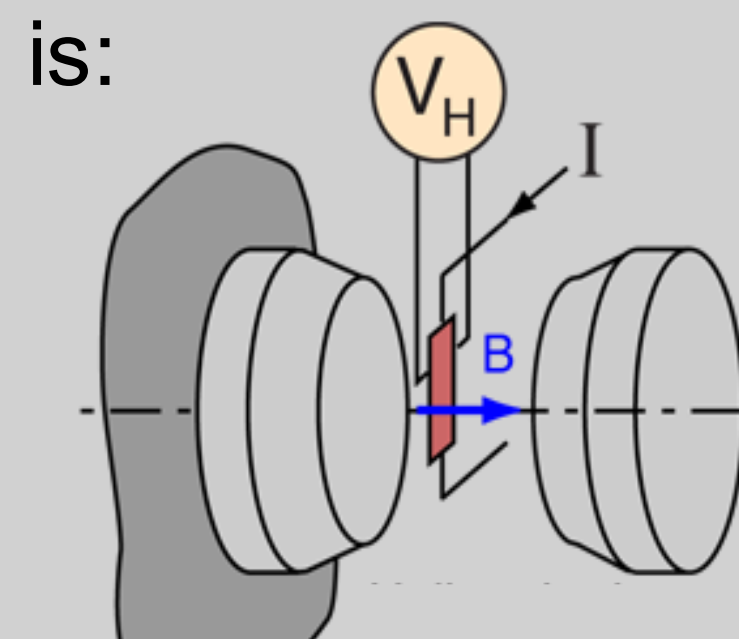
Objectives

The B field of dipole magnet has been analysed through Ansys Maxwell and optimised to create an optimal spread of bending radius for the electron beam. However, the equipment has not been characterized. The final goal of this project is to experimentally verify the modelled magnetic field data, which is executed through several milestones:

1. Use Hall effect sensor for reliable B field measurements
2. Set up a stage for 3-dimensional motion and mount sensor
3. Develop an automatic data collection, processing, and visualization pipeline
4. Scan the complete dipole B field
5. Compare measurement with simulation results

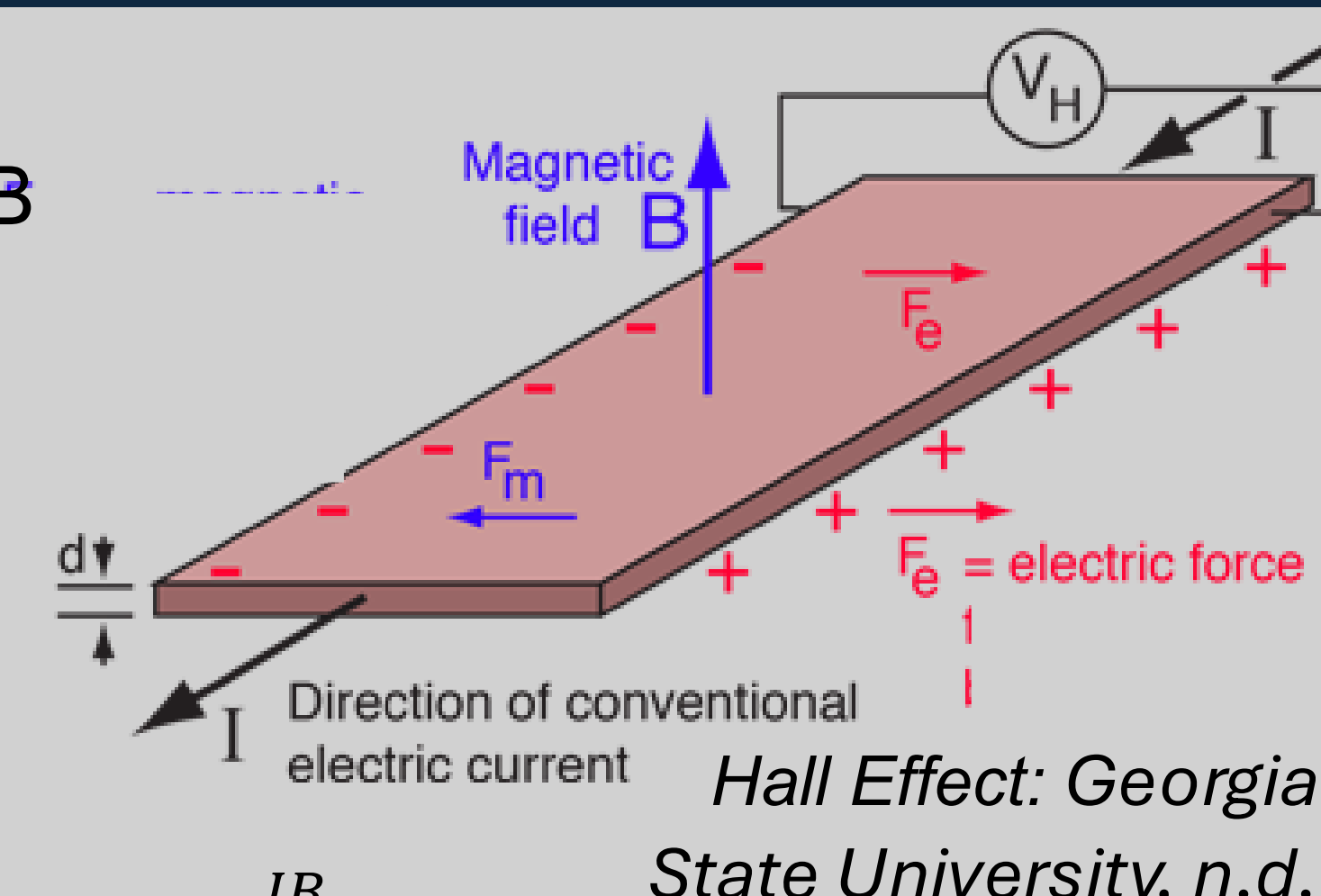
Hall Effect Sensor

Hall effect occurs when a current-carrying conductor is placed in a magnetic field (B): the B field generates a Lorentz force that pulls negative charge carriers, electrons, to one side. This separation of charge creates an electric field across the conductor to balance the magnetic force. The resulting voltage difference across the conductor is the **Hall voltage**, which is:



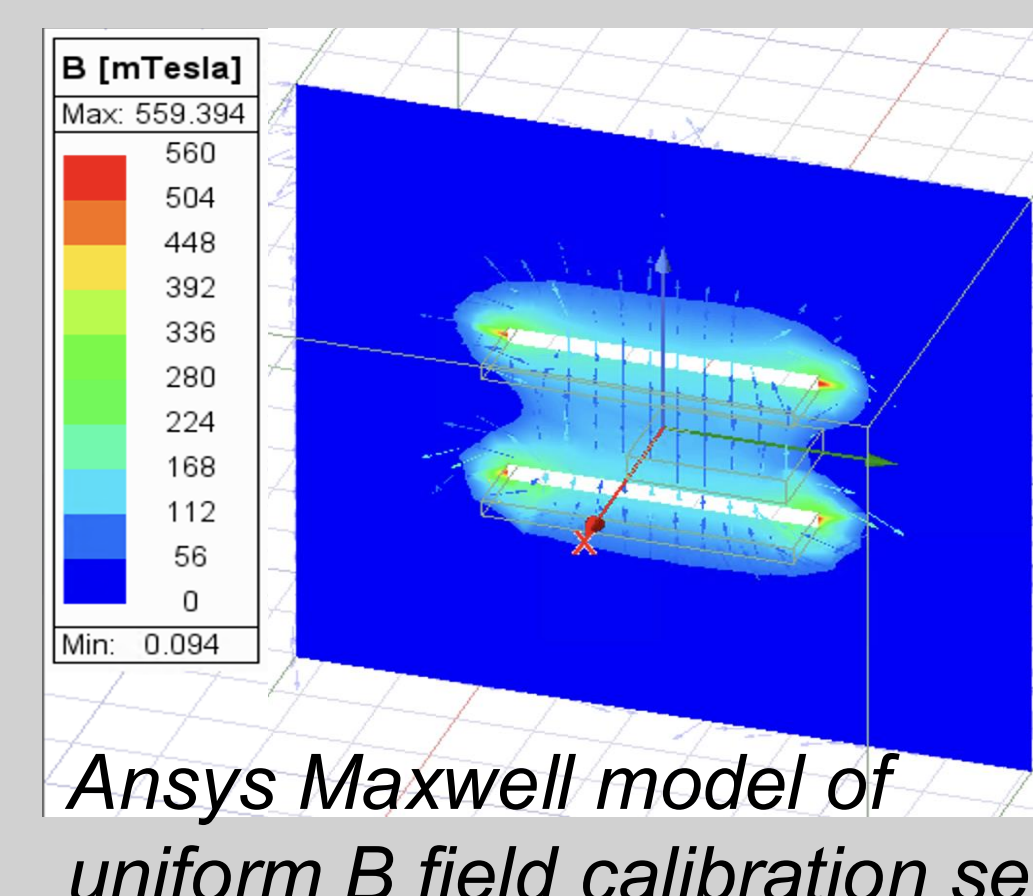
Hall Effect Sensor: Georgia State University, n.d.

where l is the height of the conductor, I is the current, n is the charge carrier density, and e is the electron charge. The Hall effect sensor has a conducting sheet in an integrated circuit with a known current. It measures the Hall voltage determine the external magnetic field strength.

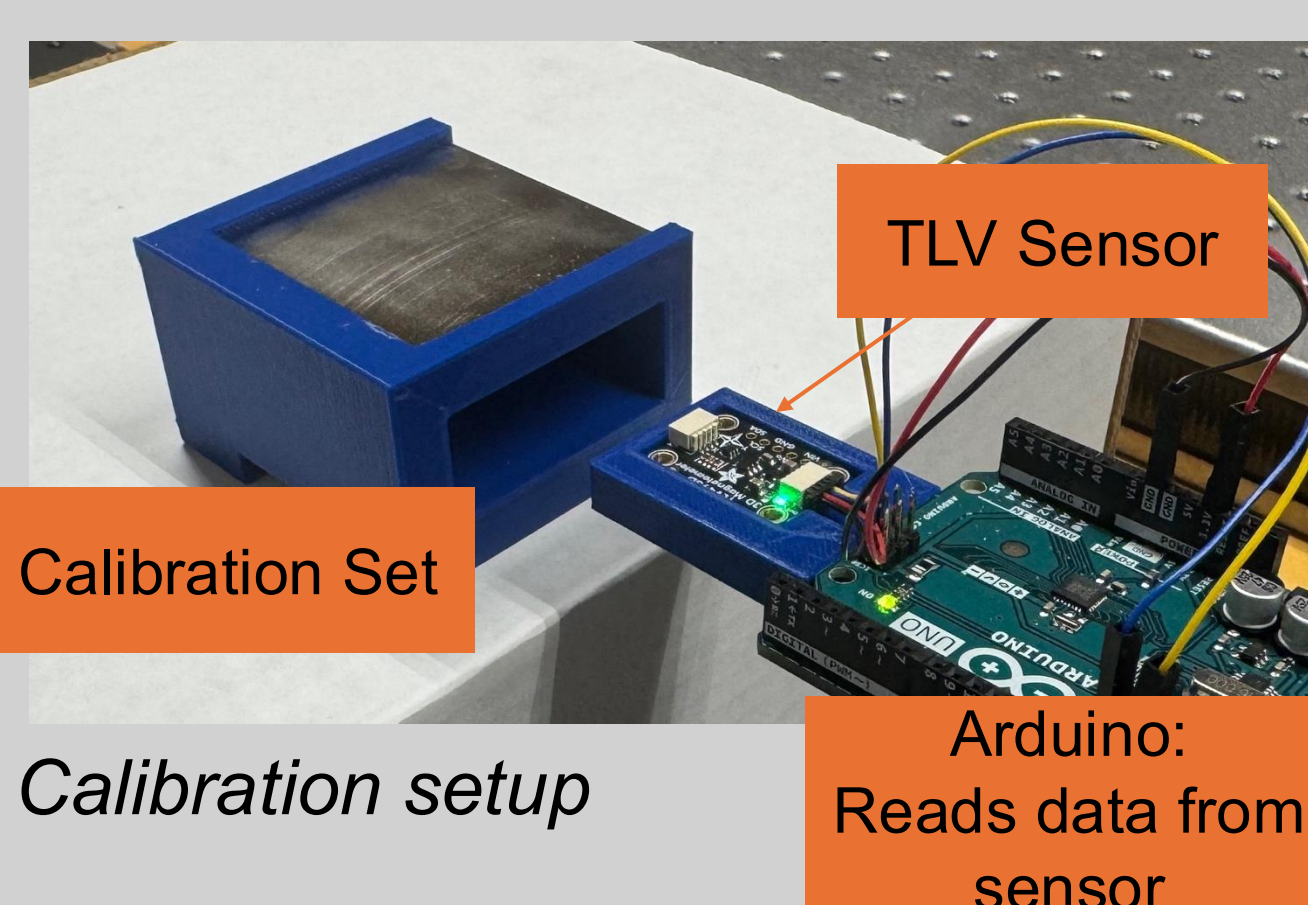


Hall Effect: Georgia State University, n.d.

We chose a commercial Adafruit TLV493D Triple-Axis Magnetometer. To calibrate it, we first attempted a sanity check against a gaussmeter in an approximate dipole field. Many inaccuracies in the setup, including the precise magnetic field, near-field approximation, and sensor positioning made this unreliable. Therefore, we built a calibration set with two parallel magnets forming a uniform B field.



Ansys Maxwell model of uniform B field calibration set



Calibration Set

The field was optimised through Ansys Maxwell by modelling different magnet dimensions and gap sizes. The final design uses two 2*2*1/8-inch Neodymium magnets separated by a 1-inch gap. The calibration holder was CAD-designed and 3D-printed with plastic to avoid interfering with the field. The simulated uniform field was 88.48 mT, and the TLV493D measured 88.19-29 mT when fixed at the set center, which confirms its reliability.

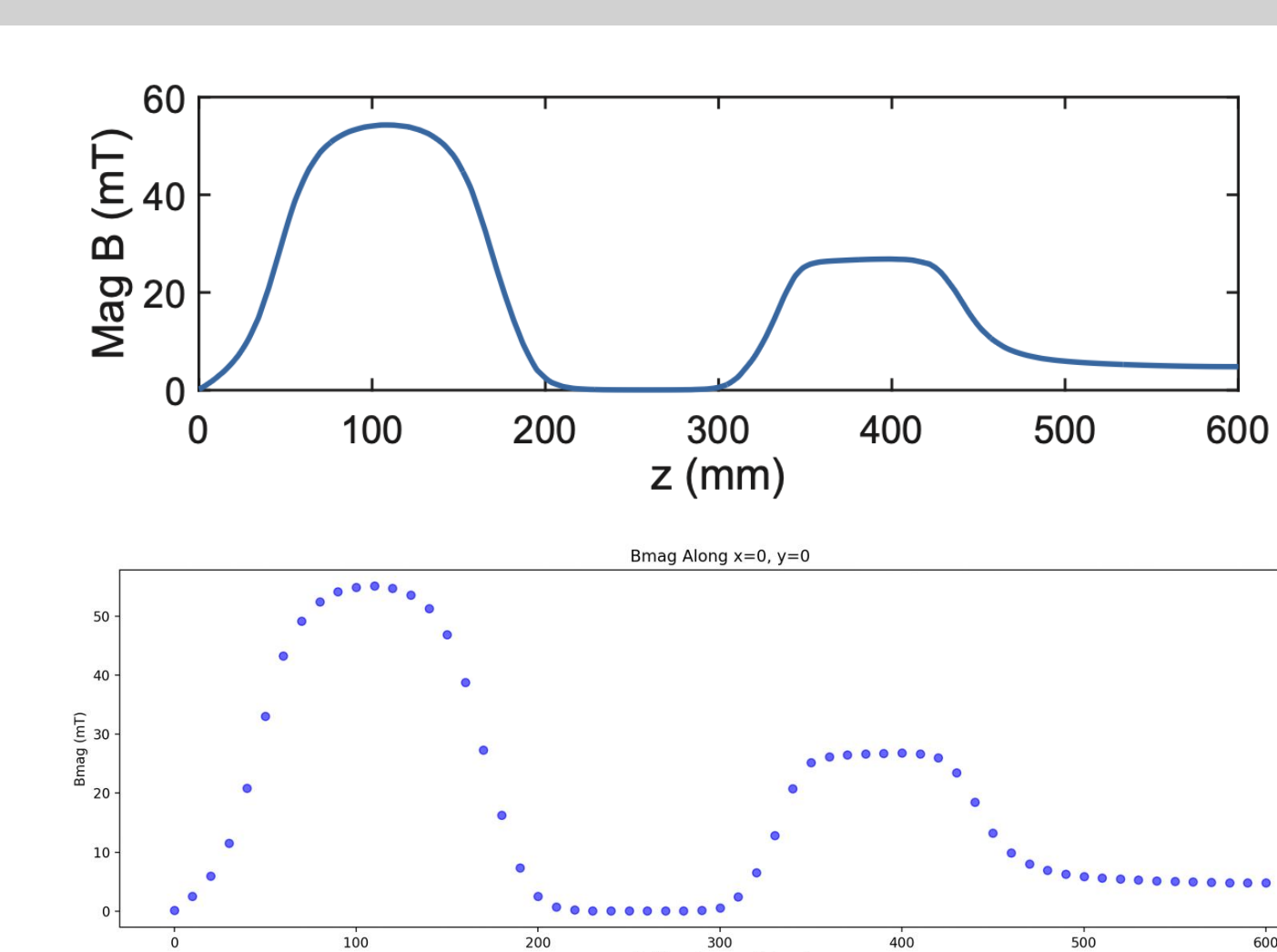
Data Pipeline

The sensor is controlled via a Python data pipeline:

Collection: It automates motor stepping and field measurement at each point. Raw measurements are saved as timestamped JSON files, which has become the gold standard for open-source data.

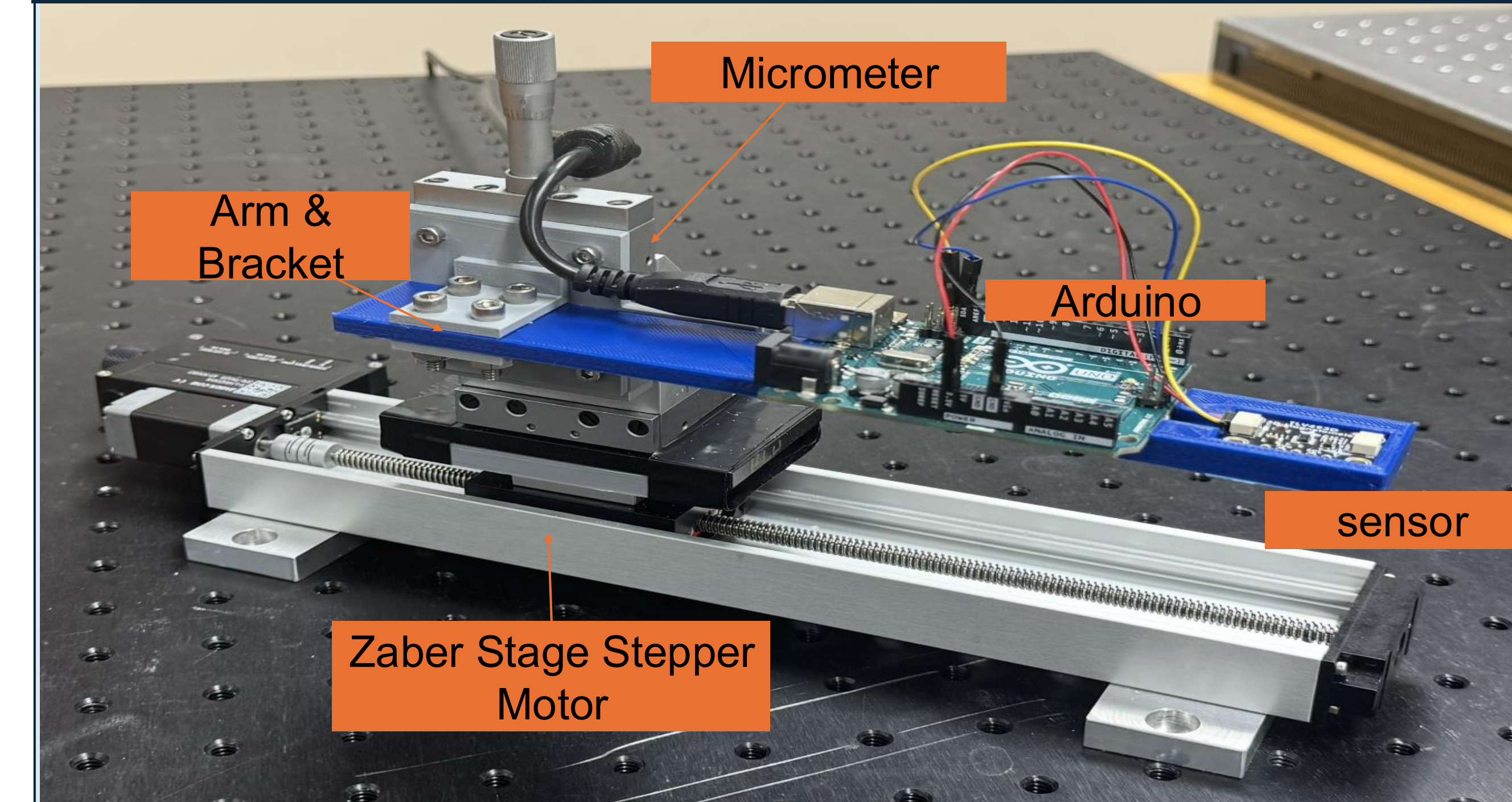
Processing: Raw three-axis B field readings are field magnitude. And motor-controlled offsets are normalised as spatial coordinates.

Visualization: Processed data is rendered to 1D scatter plots along user-specified planes and axes for analysis.



Top: Ansys Maxwell B Field result along beam axis through focusing solenoid and dipole pole pieces, which is the second smaller bump;
Bottom: Python visualisation with exported data

Hardware Setup

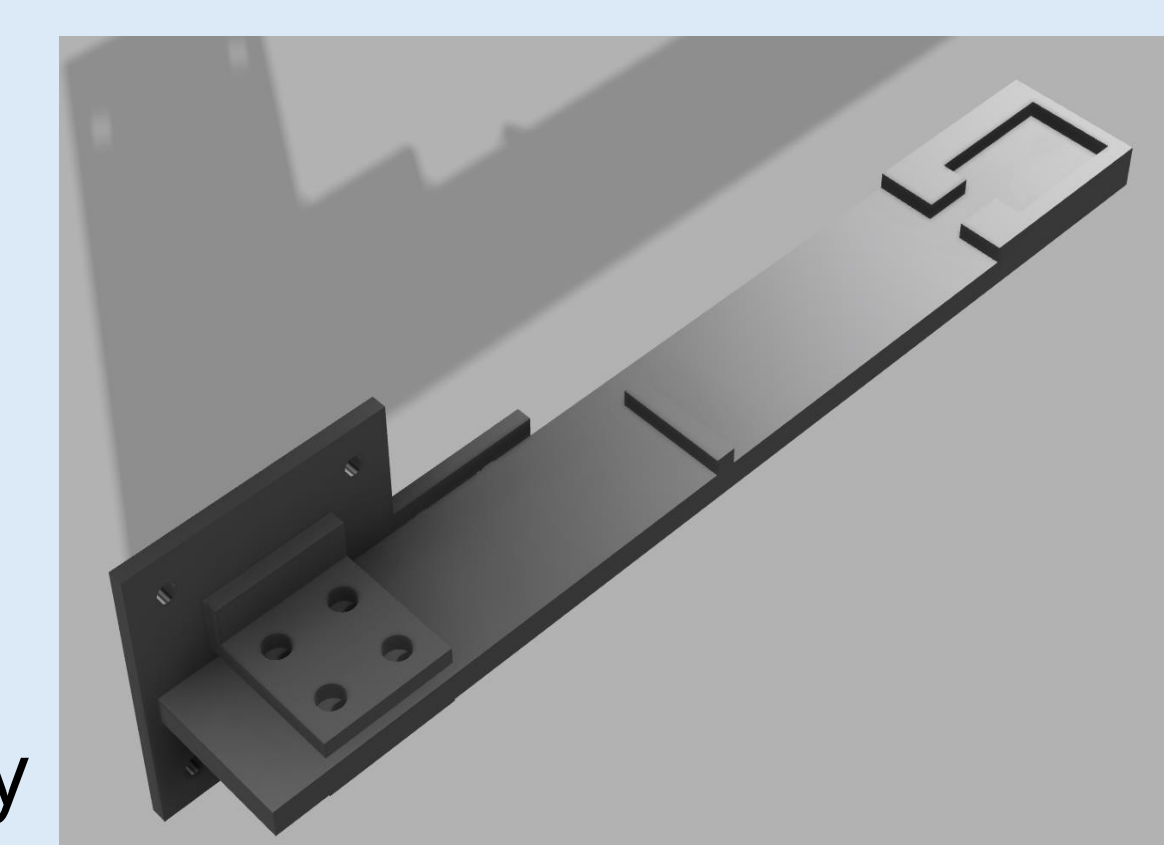


Measurement hardware setup

To control and move the sensor, the full hardware setup for measurement includes:

- **Arduino microcontroller** is compatible with TLV493D sensor. We programmed it to write the 3-dimensional field data from the sensor to serial upon request.
- **Zaber stage X-LHM200A Stepper Motor** provides controlled motion along one axis. The stepper motor has a range of 0-20cm and can be controlled on a nm scale through programming, which is sufficient across even the longest diagonal axis across the pole pieces.
- **Micrometers** allow fine-grain manual adjustment of 0-5cm along two other axes to achieve 3D motion.
- To fully scan the 3D field, we automatically step the zaber stage and manually offsets the other two axes with micrometers.

- **Stages and Brackets** are CAD-designed and 3D printed. Besides the interlocking parts, the weight handling 20cm arm holding out the sensor was the most challenging. We optimised the printing efficiency and rigidity of PLA materials to print at the infill level of 50%.



CAD model for Bracket and Arm, which are screwed together

Future Steps

With the field measurement instrument designed, built, and calibrated, we are currently waiting to open the detection chamber for further measurements with the dipole. The detection chamber is currently pumped down to near atmospheric pressure. We aim to pursue the following in the future:

- Measure data from actual dipole field
- Compare measured field to analytical results

Acknowledgements

Thank you to, **Dr. Samantha Lewis**, for her guidance and support throughout the summer, the **Wellesley College Science Center Summer Research Program** for this opportunity, and to **Dr. Katie Hall** and **Knapp Center** for supporting the equipment and resources.

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- [2] S. Lewis, A THz-Driven Electron Gun. PhD thesis, University of California, Berkeley 2020